

Lab Experiment-III: Jet Impingement

Objectives

- (i) Study the force generated by a liquid jet impinging on a surface by adjusting a balancing weight.
- (ii) Observe the variations in the force for different jet speeds.
- (iii) Interpret the results using the Reynolds transport theorem.

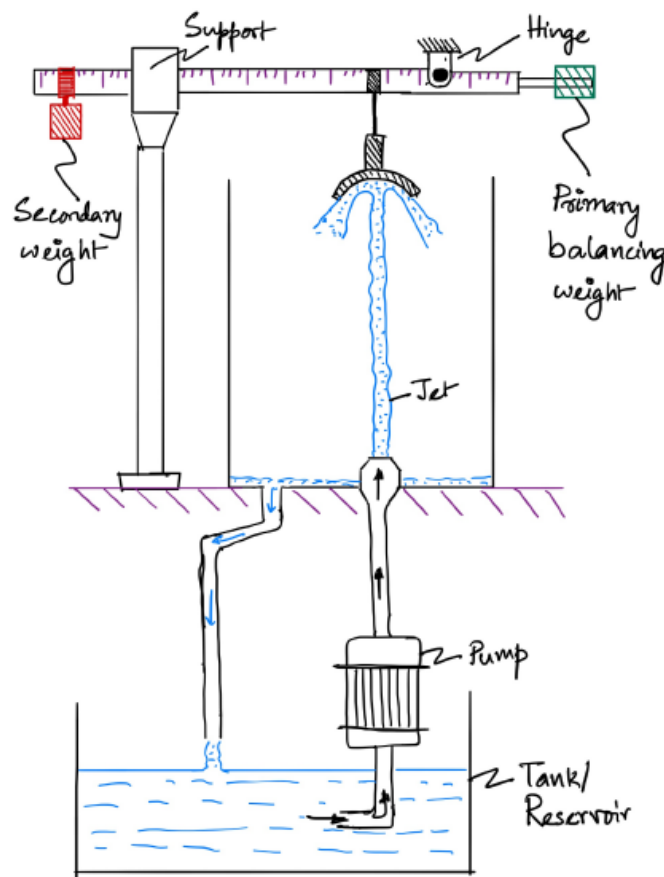
The Essential Background

1. Why are liquid jets important? Briefly discuss any 3 applications of liquid jets.
2. Read up on the Reynolds transport theorem (RTT). How is momentum conservation expressed in this theorem?
3. Can you compute the force exerted by a liquid jet impinging on a flat surface, using the RTT?

Task 3.1: In the report for Lab Exp.-III, concise answers to all the above questions have to be submitted.



(a)



(b)

Figure 3: Schematic for the experiment on jet impingement.

About the experiment

Figure 4(a) depicts the photograph of a typical jet impingement experimental set up and figure 4(b) demonstrates a schematic of the set up with various components. The experimental setup primarily consists of a nozzle through which a water jet emerges vertically in such a way that it may be conveniently observed through the transparent sheet. It strikes the target vane positioned above it. The force applied on the vane by the jet can be measured by applying balancing weights to counteract as a reaction force for the impact of the jet. Vanes are interchangeable i.e. flat plate & hemispherical. Arrangement is made for the movement of the plate under the action of the jet and also because of the weight placed on the loading pan. A scale is provided for carrying the vanes to its original position i.e. as before the jet strikes the vane. The setup is connected to a Hydraulic Bench with a flexible pipe line.

The experiment will be explained during the lab classes. Carry out the steps carefully with help from the TAs and the lab staff and record the experimental observations. A lab manual with the necessary data will be provided in Google Classroom.

Task 3.2:

- (a) Measure the balancing weight and its position for different flow rates. From this calculate the force exerted on the vane by the liquid jet.
- (b) Compute the theoretical force using the RTT. Compare the measured force and the theoretically calculated force.
- (c) Report experimental uncertainties and quantify errors by running the experiments multiple times.
- (d) In case there are mismatches between observed and the calculated forces, can you specify some reasons which may cause these mismatches?

Tasks 3.1 and 3.2 should be included in the report for this experiment.